

Deep-Q-Network in Multiview Ensemble Learning to Predict Anti-Diabetic Peptide and Diabetic Types

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Abstract—Diabetes mellitus disease has become a significant human health concern globally, causing cardiovascular complications, kidney disease, cerebrovascular accidents, etc. In humans, bioactive molecules, specifically Antidiabetic peptides (ADPs), target β or T-cells to regulate insulin production, making them potential alternatives for diabetes therapy. Recently, several computational ensemble methods have been investigated to predict ADPs. Notwithstanding progress in ensemble techniques, finding optimal solutions for predicting ADPs requires continuous effort for improvement. Therefore, the current study proposed a novel Deep-Q-Network (DQN)—based Reinforcement learning (RL) approach for optimizing a multiview ensemble to predict anti-diabetic peptides and diabetic types. Initially, to obtain the multiview, diverse feature encodings were employed, which were then systematically categorized into three heterogeneous groups based on the information sources and information-extracting techniques. Each group is considered as a view, and these views are based on sequence, physicochemical properties with sequence composition, and evolutionary information. Next, the proposed ensemble is constructed with 36 branch classifiers derived from 6 convolutional neural networks (CNNs). A reinforcement learning technique is employed to optimize the ensemble search space of $2^{36} - 1$, made by thirty-six branch CNNs and three views. The DQN approach encodes 36-bit binary strings to find the optimal ensemble. The optimal ensemble with an ideal view-classifier pair enhances antidiabetic prediction efficacy. It surpasses existing state-of-the-art with a classification accuracy of 96.37%, a Matthews correlation coefficient of 0.870, and an area under the curve of 0.983 on the standard ADPs dataset. Lastly, the performance of the proposed ensemble is considered to be its efficacy in peptide antidiabetic activity prediction.

Index Terms—Antidiabetic peptides (ADPs), deep-Q-network reinforcement learning, deep learning, multiview, optimal ensemble.

I. INTRODUCTION

DIABETES mellitus (DM), often known as diabetes, is a chronic disease that has a detrimental effect on human health [1]. Specifically, diabetes disease upregulates blood glucose levels. It happens when the human body either fails to produce sufficient insulin or cannot utilize the insulin it generates. Diabetes appears in two types: insulin-dependent (Type-1) and non-insulin-dependent (Type-2) [2]. Type-1 diabetes (T1D) is an autoimmune disorder wherein the immune system erroneously attacks and obliterates the cells that generate insulin

in the pancreas. Conversely, Type-2 diabetes (T2D) is the most common form, characterized by the body's resistance to insulin or insufficient insulin production by the pancreas. This condition progressively detrimentally impacts organisms, manifesting as blurred vision, kidney disease, cardiovascular complications, poor wound healing, and numbness [3].

According to statistics from the International Diabetes Federation (IDF), as of 2021, around 537 million individuals globally are affected by diabetes [4]. This metabolic disease accounts for 6.7 million fatalities per year [5]. Despite advancements in the medication field still, there is no cure for DM [6]. Traditional insulin injections are generally used for DM management but result in several problems, including hyperinsulinemia, discomfort, and soreness. Since the existing therapies have specific limitations, there is an urgent necessity to create novel anti-diabetic therapies for diabetes.

Many bioactive peptides demonstrate anti-diabetic characteristics and are a promising alternative for diabetes treatments [7]. Antidiabetic peptides (ADPs), components of the host defense peptides (HDPs), demonstrate significant potential in regulating glucose metabolism [8]. In drug discovery, the application of ADPs offers additional benefits over conventional non-peptide medications. The benefits include high specificity, less susceptibility to drug resistance, and easy modification and synthesis [9].

The task of anti-diabetic activity prediction can be formulated as a binary classification problem. To classify ADPs, a set of n peptides is represented by feature vectors $X = \{x_1, x_2, \dots, x_n\}$, where each $x_i \in \mathbb{R}^d$ and d denotes the dimensionality of the feature space. Correspondingly, each peptide has a label $y_i \in Y$, where $Y \in \{-1, 1\}$. If $y_i = -1$, it indicates that the peptide is non-antidiabetic, and if $y_i = 1$, it is antidiabetic. In ADP classification, the aim is to learn the classification function, which can be expressed mathematically as $f(X) \rightarrow Y$. The prediction function $f(x)$ is a classifier that classifies the antidiabetic activity utilizing the label $\{-1, 1\}$.

Recently, some studies have investigated the peptides possessing anti-diabetic activity utilizing a computational approach. AntiDMPpred [10] built a computational model utilizing composition-based features. The drawback of this research is the absence of an evaluation of optimal multiview feature combinations. ADP-Fuse [11] developed a web server that employs multiview features to predict antidiabetic activity and its type. STADIP [8] proposed a stacking ensemble with a 2-step feature selection to predict ADPs, etc. Although the ADP-Fuse [11], STADIP [8] used an ensemble, but the limitation of an ensemble is that not all classifiers enhance ensemble performance [12].

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Thus, identifying the optimal ensemble is crucial [13]. The optimal ensemble seeks to determine the subset classifiers $S \subseteq C$ that maximizes the ensemble performance $\mathfrak{P}$ (e.g., AUC, accuracy), where $C = \{C_1, C_2, \dots, C_n\}$ denotes the ensemble pool of n classifiers. The optimization process aggregates only best-performing individual classifiers' contributions $\mathfrak{P}(C_i)$ in the ensemble design, which can be mathematically represented as $\max_{x \in \{0,1\}^n} \sum_{i=1}^n w_i \cdot x_i \cdot \mathfrak{P}(C_i)$. Where $x_i \in \{0,1\}$, $i = 1, 2, \dots, n$ is a binary vector that denotes the inclusion or removal of each classifier, and w_i is a weighting coefficient associated with each classifier. This optimization problem is also subject to constraints $L \leq \sum_{i=1}^{100} x_i \leq U$, enforcing a lower and upper bound on the cardinality of the subset.

Moreover, with swift advancement in AI, multiview learning techniques have evolved in the domains of data analytics. In peptide structure or function prediction, multiview learning enables the capture of complex patterns in peptides, improving model predictive performance [14]. Multiview integrates different viewpoints and can be obtained through distinct feature sets (views) $V = \{v_1, v_2, v_3, \dots, v_n\}$, where each views V_i gives complementary knowledge [15]. The multiview learning aids to tackle the challenges in machine learning that emerge when multiple feature sets are employed to describe information [16]. The aim of multiview learning is to optimize the aggregate loss across all views, expressed as $\min \sum_{v=1}^n L(f(v_n), y)$, to reduce generalization error and ensure consistent performance, even if one view is less informative or noisy.

In recent years, substantial advancements have occurred in the domain of multiview and ensemble learning. Nevertheless, identifying the optimal combination of view-classifier pairs for the ensemble remains a challenge, which can be efficiently addressed by reinforcement learning (RL). Therefore, this study proposed an RL-based Deep-Q-Network (DQN) to optimize a multiview deep ensemble for two-level prediction (i.e., first classifying peptides as antidiabetic vs. non-antidiabetic, and subsequently distinguishing between T1D and T2D ADPs). The proposed model is an optimized multi-branch $1 - d$ CNN ensemble built by 36 branch-classifiers derived from 6 CNNs and RL employed for ensemble optimization. The proposed ensemble utilizes two CNN architectures for the employed six CNNs: the first architecture is used for the three CNNs, while the second architecture is for the remaining three CNNs. In this study, to achieve a multiview model, seven feature encodings were used, which were then systematically categorized into three heterogeneous feature groups based on the information sources and information-extracting techniques. Each group is considered as a view, and these views are based on sequence, physicochemical properties with sequence composition, and evolutionary view. Next, to predict the antidiabetic activity, we employ an ensemble classifier and all three views. Constructing an optimal ensemble is challenging since not every classifier in an ensemble contributes equally to performance improvement [17]. Additionally, there are different classifiers and encoding techniques exist, but the best approach is to use both at once. In the proposed ensemble, 36 view-classifier pairs exist (36 branch-classifiers derived from 6 CNNs), resulting in a total of $2^{36} - 1$ ensemble combinations possible. Identifying the

optimal view-classifier combination from a large search space is challenging. The problem of large search space ($2^{36} - 1$) can be effectively addressed through the DQN [18]. DQN uses a neural network to approximate the Q-value, making it scalable for large state-action spaces, where conventional Q-learning is infeasible due to the requirement for a massive Q-table 2^{36} ($2^{\text{Possible_states}}$).

In the present work, the proposed model utilizes a fusion of DQN-based RL with deep ensemble to achieve the desired goal. Here, RL aims to identify the best view-classifier combination within a vast search space. In addition, the number of branch classifiers in the proposed ensemble is not fixed. Instead, we represent the ensemble by selecting the best view-classifier combination using the RL. Although no effort was given to create an optimal ensemble to predict antidiabetic activity, extensive experimentation validated the efficacy of the proposed ensemble classifier. Thus, the contributions of the work can be summarized as follows:

- 1) To our knowledge, the proposed model is the first to explore reinforcement learning to optimize multiview deep ensemble for predicting peptide anti-diabetic activity.
- 2) The proposed model utilizes DQN to approximate the Q-value in RL, making it computationally effective in finding optimal ensemble in a large search space.
- 3) The predictive efficacy of the proposed ensemble is examined using the training and test sets, and the findings demonstrate that the model has considerable potential for predicting antidiabetic activity.
- 4) Extensive investigation of standard datasets and statistical analysis utilizing multiple views validates the proposed ensemble classifier's superiority over existing state-of-the-art approaches.

The following portions of the article are organized as follows. Section II reviews previous studies. Section III discusses the proposed methodology. Section IV presents the experimental setup and findings. Section V discusses the conclusion.

II. RELATED WORK

Diabetes presents significant challenges due to its chronic effects on human health. ADP peptides have demonstrated considerable efficacy in managing and alleviating diabetes-related symptoms. Consequently, several advanced computational techniques have been utilized to accurately predict ADPs, aiding in the development of novel treatment alternatives. Chen et al. (2022) [10] developed the first AntiDMPpred web server for predicting ADPs from experimentally confirmed samples. AntiDMPpred utilized an ML-based random forest classifier with 5-fold validation and four feature encoders for ADP prediction. This model achieves an accuracy of 77.12% and an area under the curve (AUC) of 0.8193. The main limitation of this study is that the negative dataset is created by aggregating peptides from AVPdP, which exhibited the lowest likelihood of non-ADPs. Subsequently, ADP-Fuse [11] also utilizes eight ML classifiers to predict ADPs and their type. ADP-Fuse exhibits Mathew's correlation coefficient (MCC) of 0.841 and an accuracy of 0.940, indicating satisfactory performance; yet, it requires

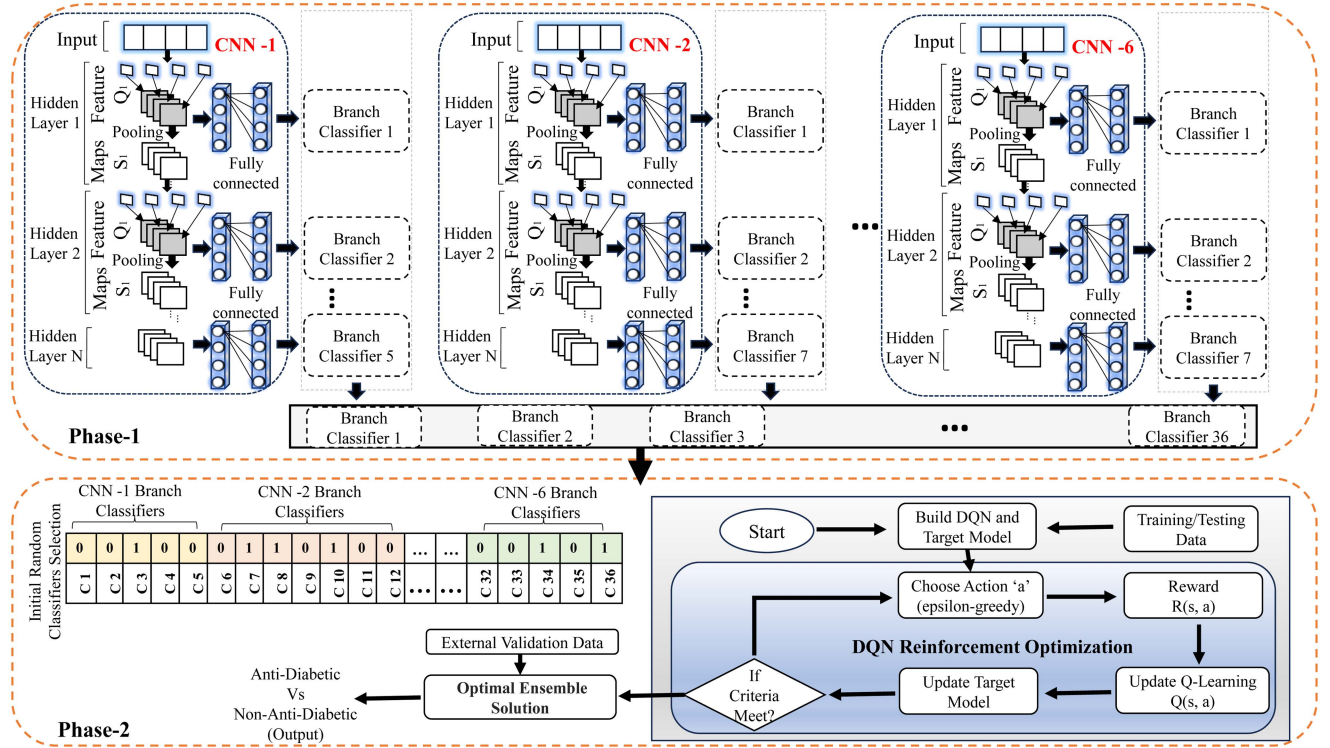


Fig. 1. Workflow of proposed model: Phase (1) a multi-branch CNN ensemble construction with multiview learning for ADPs prediction, Phase (2) DQN to determine the optimal view-classifier pair for ensemble formation.

improvement as it remains below average. Most recently, a stacking ensemble called STADIP [8] was used for ADP prediction. It uses 12 feature encodings and 7 ML classifiers. It employed two-step feature selection to boost the predictive performance and got an accuracy and MCC of 0.954 and 0.877, respectively. This study solely predicts ADPs and does not attempt to classify their types. Similarly, another model, BertADP [19] utilizes large-scale pre-trained protein language models to predict ADPs, but performed experiments with a 110-sample subset. AntiT2DMP-Pred [20] utilizes feature fusion with ML to predict type-2 diabetes and achieves 0.957 and 0.851 accuracy and MCC, respectively. Note: both BertADP [19] and AntiT2DMP-Pred [20] utilize different datasets.

Moreover, a generative model called SeqGAN was employed to discover the new ADP sample [21]. Like ADP-Fused, it utilizes the same ADPs from the BioDADPep [22] database while utilizing other non-ADPs. It uses the evolutionary scale model for preprocessing and shows 90.48% accuracy with the CNN classifier. In diabetes drug discovery, a graph convolutional network called IP-GCN [23] was built to predict insulin and non-insulin proteins. Apart from antidiabetic research, additional peptide-based studies have also been conducted, including Gan-based augmentation, to predict anti-viral and anti-coronavirus peptides [24], Peptide. The PACVP [25] predicts anti-coronavirus peptides employing stacking learning. Thus far, the DM has impacted millions and caused deaths. Further research is necessary due to its chronically harmful nature. Notwithstanding the progress achieved, there remains a significant opportunity for substantial enhancement in DM therapeutics and pharmacology.

TABLE I
DETAILS OF BENCHMARK ADP-FUSE DATASETS

Dataset	Positive	Negative
ADP-Fuse ₁ ^{Training}	658▲ and 600★ (ADPs)	658★ (non-ADPs)
ADP-Fuse ₁ ^{Independent}	219▲ (ADPs)	741▲ (non-ADPs)
ADP-Fuse ₂ ^{Training}	500▲ and 432★ (T1D)	200▲ and 168★ (T2D)
ADP-Fuse ₂ ^{Independent}	155★ (T1D)	22▲ (T2D)

▲-Without CD-HIT, ★-after CD-HIT

III. METHODOLOGY

The proposed workflow of antidiabetic activity prediction is illustrated in Fig. 1. The methodology section is discussed in three subsections: Section III-A provides a detailed overview of the dataset. Section III-B: discusses feature encoding and multiview learning. Section III-C describes the detailed working of the proposed method.

A. Dataset

A series of experiments was conducted to evaluate the performance of deep-learning models in predicting anti-diabetic activity. For experimental purposes, the current investigation utilizes four benchmark anti-diabetic ADP-Fuse [11] datasets, as Table I outlines. ADP-Fuse collected experimentally validated anti-diabetic peptides from the BioDADPep [22] database. Conversely, peptides without anti-diabetic properties are gathered from prior other studies [26], [27]. These non-ADPs include random peptides and those peptides that demonstrate antibacterial action, such as anti-hypertensive and anticancer properties.

TABLE II
AMINO ACID-BASED FEATURE

Feature	View	Feature Vector
AAC	Sequence-Based (SB)	20- d
DC		400- d
SPP	Physicochemical Properties & Sequence Composition (PPSC)	15- d
PseAAC		(20 + α)- d
QSO		46- d
PAM250	Evolutionary Based (EB)	75- d
BLOSUM62		75- d

Among the four datasets utilized, two are used for training purposes, and two are independent datasets for external validation. To predict peptide anti-diabetic activity, this study employed the $ADP - Fuse_1^{Training}$ and $ADP - Fuse_1^{Independent}$ datasets. Additionally, the ADP sample from the first training and independent datasets includes both T1D and T2D peptides. Based on previous research, this training dataset is partitioned into 80:20 ratios for training and validation purposes [11]. Next, to assess the model's generalization capability and confirm its effectiveness on unseen data, the $ADP - Fuse_1^{Independent}$ dataset is used. Prior studies, such as ADP-Fuse [11], removed identical samples but did not employ CD-HIT for ADPs because a high threshold obtained fewer samples. This small sample size may not be enough for a reliable model [28]. In the present study, we also removed identical samples; additionally, the CD-HIT impact was evaluated using a 90% threshold. Upon applying CD-HIT to the $ADP - Fuse_1^{Training}$ data, we obtained 600 ADPs. But, for non-ADPs, we used CD-HIT [29] with a 0.6 threshold to remove redundant negative samples.

Moreover, to identify the diabetic type, this study employed $ADP - Fuse_2^{Training}$ and $ADP - Fuse_2^{Independent}$ T1D and T2D peptides datasets. The dataset $ADP - Fuse_2^{Training}$ is used to train the model, while $ADP - Fuse_2^{Independent}$ is used for external validation. These T1D and T2D peptides are identical to those used in the $ADP - Fuse_1^{Training}$ and $ADP - Fuse_1^{Independent}$ datasets. Like the first dataset, we also applied CD-HIT (90% threshold) to the second dataset. Upon applying CD-HIT to the $ADP - Fuse_2^{Training}$ data, we obtained 432 T1D and 168 T2D peptides. Additionally, the length of the peptide in all datasets ranged from 7 to 40 characters. Lastly, the limitations of this dataset are that it has a relatively smaller size of T2D peptides (imbalance data), but the ensemble framework effectively handles it. These datasets can be freely downloaded (from the ADP-Fuse web) [11].

B. Feature Encoding and Multiview Learning

This section presents the seven amino acid encoding techniques [30] employed in the current study. As discussed in Section I, to attain multiview, we organized encoding techniques into 3 heterogeneous sets, each set considered a distinct view. These views are centered on sequence, physicochemical properties with sequence composition, and evolutionary features. The summary of used features is presented in Table II.

1) *Amino Acid Sequence Based Feature*: This subsection utilizes two benchmark sequence-based feature descriptors from

prior studies for the current study: Amino acid composition(AAC) and Di-peptide Composition(DC).

AAC encoding is employed by numerous studies to predict anti-diabetic peptides [11], anti-coronavirus [25], anticancer [31], and more. AAC calculates the ratio of each amino acid within the peptide sequence and generates a 20-d feature vector, which can be expressed as

$$AAC_{feature} = \left(\frac{n_{A_1}}{L}, \frac{n_{A_2}}{L}, \dots, \frac{n_{A_{20}}}{L} \right) \quad (1)$$

where $AAC_{feature}$ is the peptide feature vector representing the proportion of each of the 20 amino acids in the peptide sequence, n_{A_i} represents the number of occurrences of amino acid A_i , and L is the peptide length. Additionally, another descriptor is DC. Like AAC, DC has also been used in prior studies to predict anti-diabetic [11] peptide, and protein malonylation sites [32]. DC represents the frequency of amino acid pair occurrences in a sequence. There are 400 pairwise combinations possible for 20 amino acids, generating a 400 - d feature vector. For the peptide of length L . DC can be mathematically expressed as

$$DC(A_i, A_j) = \frac{N_{ij}}{L - 1} \quad (2)$$

where (A_i, A_j) denotes a pair of amino acids, $N_{i,j}$ indicates the dipeptide frequency (A_i, A_j) within the sequence.

2) *Physicochemical Properties With Sequence Composition*: The physicochemical properties and sequence composition of amino acids are vital for developing protein function and structure. This subsection incorporates three descriptors derived from prior studies: Standard Physicochemical Properties (SSP), Pseudo Amino Acid Composition (PseAAC), and Quasi Sequence Order (QSO).

SPP refers to the intrinsic physical and chemical properties that affect peptide structure, behavior, and interactions. Aliphaticity, hydrophilicity, Hydrophobicity, charges, atomic composition, etc., are SPP features [30]. Based on earlier studies [31], a total of 15 SPP are used, which generate a 15 - d feature vector. Subsequently, the second employed descriptor is PseAAC, which has previously been utilized to predict anti-diabetic [11], and anticancer [33] peptides. It integrates the AAC with additional sequence-order details, capturing the interrelations among residues and their physicochemical properties, such as hydrophilicity and hydrophobicity, to produce an improved, meaningful representation of peptides. For the peptide sequence $P_{Sequence} = (\rho_1, \rho_2, \rho_3, \dots, \rho_n)$ where $i \in \{1, 2, \dots, n\}$ and ρ_i denotes the i -th residue of the peptide, n represents the total count of amino acid in sequence; the PseAAC can be represented as $PseAAC_{feature} = \{y_1, y_2, y_3, \dots, y_{20}, y_{20+1}, \dots, y_{20+\alpha}\}$. Where y_q is defined as

$$y_q = \begin{cases} \frac{f_q}{\sum_{i=1}^{20} \frac{f_i + w}{w \cdot \tau_j} \sum_{i=1}^{\alpha} \tau_j}, & : (1 \leq q \leq 20) \\ \frac{f_q}{\sum_{i=1}^{20} \frac{f_i + w}{w \cdot \tau_j} \sum_{i=j}^{\alpha} \tau_j}, & : (20+1 \leq q \leq 20+\alpha) \end{cases} \quad (3)$$

where f_q represents the ratio of occurrences of amino acid q in a peptide P , the first 20 components indicate the AAC effect, whereas the remaining components from $(20+1)$ to $(20+\alpha)$ represent the sequence order effect, w denotes a weight factor (0.05 by default). The value of α must be shorter than the

shortest peptide length within the dataset. The upper limit of α is 30. In our employed dataset, peptide length ranges from 7 to 40 amino acids; so, $\alpha=6(\alpha \leq 7)$ and τ_j is the j^{th} tire's correlation factor, calculated as

$$\tau_j = \frac{1}{l-j} \sum_{i=1}^{l-j} (n(i) - n(i+j))^2 \quad (4)$$

where l is the peptide length, $n(i)$ indicates the numerical value of the chosen physicochemical properties (PP) for the amino acid at i^{th} position, $n(i+j)$ indicates the same amino acid's property, positions ahead of $n(i)$ and summation determines the difference between the PP of amino acids separated by j^{th} positions. Furthermore, another used descriptor is QSO. This encoding is based on the physicochemical distance among amino acids [30] and is utilized in anti-diabetic [11] peptide classification. The features of QSO are derived from a distance matrix encompassing 20 amino acids. It produces a $46-d$ feature vector, and can be mathematically defined as:

$$QSO = \begin{cases} \frac{f_q}{\sum_{i=1}^{20} f_q + w \sum_{d=1}^{nlag} \tau_d}, & : (1 \leq q \leq 20) \\ \frac{w \cdot \tau_{q-20}}{\sum_{i=1}^{20} f_q + w \sum_{d=1}^{nlag} \tau_d}, & : (d=21, 22, \dots, (20+nlag)) \end{cases} \quad (5)$$

Where f_q represents the normalized frequency of amino acid type q , w (0.05 by default) denotes a weighting factor, and $nlag$ represents the maximum possible value of the lag calculated as $nlag = l_{min} - 1$. Where l_{min} is the shortest peptide length within the dataset.

3) *Evolutionary-Based Feature*: The evolutionary technique gives numerical values to each amino acid according to substitution matrices, where each entry indicates the probability of one residue being replaced by another [34]. These values reflect the evolutionary resemblance among residues in peptides [35]. Based on prior studies, the present study employed the two substitution probability matrices, BLocks Substitution Matrix (BLOSUM62) and Point Accepted Mutation (PAM) 250.

BLOSUM62 employs a substitution matrix to determine the resemblance between two amino acid sequences. It primarily provides information regarding peptide evolution [35]. It is represented by a 20×20 matrix, where each entry $E(i, j)$ denotes the log-odds score for the substitution of amino acid i with amino acid j , computed as $E(i, j) = \log((f(i) \cdot f(j)) / P(i, j))$. Where $f(i)$ and $f(j)$ represent the frequencies of amino acids i and j , respectively, while $P(i, j)$ denotes the observed frequency of the amino acid pair in closely similar peptide sequences. For peptide sequences with a window size of n , it generates a $(20 \times n) - d$ feature vector. This descriptor is also employed to predict anti-diabetic activity [11], protein malonylation sites [32], etc. Moreover, another evolutionary feature is PAM250. It measures the probability of amino acid substitution during evolution, particularly for sequences that have diverged by 250 mutations. It calculated the log-odds score for the substitution of amino acid i with j as $E(i, j) = \log((P(i, j) / f(i) \cdot f(j)))$. PAM generates a $(20 \times n) - d$ feature vector and is also utilized to predict protein similarity [34].

C. Proposed Method

This section is bifurcated into two sub-sections. The first sub-section provides a concise overview of the proposed technique. The second sub-section discusses the technical aspects of the proposed optimized multiview learning for predicting anti-diabetic activities.

1) *Overview*: As illustrated by Fig. 1, the proposed method comprises two phases: first, a multi-branch CNN ensemble formation with a multiview learning technique for anti-diabetic activity prediction, and second, DQN reinforcement learning to identify the optimal view-classifier pair.

- The proposed ensemble employs six 1-dimensional CNNs and generates 36 CNN branch classifiers by connecting each hidden layer of all 6 CNNs to the output layer. In addition, these branches not only generate several classifiers for classifying anti-diabetic activity on intermediate-level features but are also trained to strengthen the gradients from deep to shallow layers.
- The second phase employs the DQN reinforcement learning to form an optimal ensemble by selecting the best view-classifier pair from a $2^{36} - 1$ possible view-classifier combinations. A maximum F1 score is utilized as a metric to select the best view-classifier pair.

2) *Proposed Multiview Ensemble*: The present work proposes an ensemble framework that utilizes three distinct views and GridSearchCV-tuned [36] CNNs to predict peptide anti-diabetic activity. The key reasons for employing CNNs are the excellent classification potential and broad adaptability in bio-medical data [37]. Likewise, the rationale for using an ensemble is its superior performance compared to individual classifiers [38]. In addition, an ensemble inherently mitigates class imbalance by aggregating diverse decision boundaries from different classifiers, thereby reducing bias toward the majority class and stabilizing predictions for the minority class without the need for explicit weighting or oversampling [39]. The proposed ensemble consists of six $1-d$ CNNs, implemented using two distinct CNN architectures: three CNNs of architecture-1 and three CNNs of architecture-2. Both CNN architecture contains a distinct number of layers. Architecture-1 has five, whereas architecture-2 uses seven hidden layers. The details of both employed CNN architectures are provided in the supplemental file. In addition, every hidden layer of all six CNNs is connected to an extra header(dense+output) layer. As depicted in Fig. 1, a total $C=36$ branch classifiers are constructed based on middle-level features by connecting each hidden layer to an additional header. These CNN branches not only create a number of classifiers for predicting anti-diabetic activity on middle-level features but are also trained to supplement the gradients from deeper to shallower layers. For effectively utilizing the multiview learning, two distinct CNN architectures (CNN-1&2) are trained on the SB, two (CNN-3&4) on PPSC, and two (CNN-5&6) on the EB view, as indicated in the view-classifier combination setup in Fig. 2

To train the proposed model, the current study utilized anti-diabetic peptide data $Data_{train} = \{(X_i, y_i)\}_{i=1}^n$, where $X_i \in \mathbb{R}^d$ the feature vector of the i -th sample with d -features and $y_i \in$

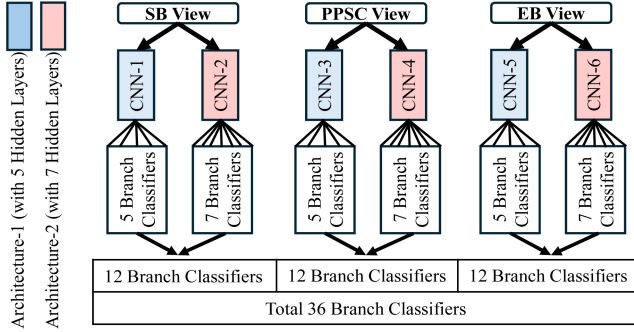


Fig. 2. Configuration of view-classifier pair in proposed ensemble classifier

$\{Anti - diabetic, Non - anti - diabetic\}$ represents the label of sample X_i . In this illustration, φ and κ denote the bias and convolution kernel term of j -th hidden layer and j -th layer weight can be calculated as $\mathbb{W}_j = (\varphi_j, \kappa_j)$, where $j \in \{1, 2, 3, \dots, J\}$ is a total number of hidden layers. The activation and convolution operation at j -th layer can be defined as

$$Z_j = f(\varphi_j * Q_{j-1} + \kappa_j), \quad (Q_0 \equiv X_i) \quad (6)$$

where Z_j denotes the feature map following the convolution and activation steps, while the symbol $*$ signifies the convolution operator, $+$ symbol adds bias term φ , and the tensor Q_{j-1} denotes the feature map at the hidden layer $j-1$. We employed ReLu $f(\mu) = \max(0, \mu)$ activation in CNN [12] because of its effective learning rate and classification performance. Afterward, the compressed feature maps Q_j are derived from the feature maps Z_j at hidden layer j by applying a max pooling as $Q_j = g(Z_j)$.

Furthermore, an extra output layer is connected to each hidden layer of all six CNNs, whereas in the sigmoid classifier, feature maps are fed through the fully connected layer. Within each branch network, to represent input vectors and weight matrix of fully connected layer $I^{(j)}$ and $\mathbb{W}_{full}^{(j)}$ terms are used respectively where $I^{(j)}$ is the flattened representation of the feature map $Z^{(j)}$, which is derived from the j -th hidden layer. Following the fully connection, the input vector $\mathbb{R}^{(j)}$ of the output layer in branch j can be derived as:

$$\mathbb{R}^{(j)} = \mathbb{W}_{full}^{(j)} \cdot I^{(j)} + \sigma_{full}^{(j)} \quad (7)$$

Where $\sigma_{full}^{(j)}$ represents the bias term used in the output function of the fully connected layer. Lastly, a sigmoid function $\mathfrak{s}^{(j)}(.)$ is appended to the end of branch j in order to construct the branch classifier j :

$$\mathfrak{s}^{(j)}(Y = b|X_i) = \frac{1}{1 + \exp(\mathbb{R}^{(j)})} \quad (8)$$

Where Y represents the potential class label of a sample X_i . For instance, in a binary classification task on anti-diabetic activity prediction, the likelihood that sample X_i is identified as anti-diabetic by branch classifier j is calculated by $\mathfrak{s}^{(j)}(Y = b|X_i)$.

Moreover, the loss function is crucial to evaluate the model performance. It measures the discrepancy between the predicted probability distribution and the actual class label, where a lower cross-entropy indicates a lesser divergence between the predicted outputs and the actual labels [12].

By incorporating both the weighted matrix and bias term of the full connection layer of branch classifier j as $\delta_{full}^{(j)} = (\mathbb{W}_{full}^{(j)}, \sigma_{full}^{(j)})$, the loss function is defined as:

$$\begin{aligned} \text{Loss}^{(j)}(\mathfrak{W}^{(j)}, \delta_{full}^{(j)}) = & - \sum_{i=1}^n \left[Y_i \log \mathfrak{s}^{(j)}(y = 1 | X_i) \right. \\ & \left. + (1 - Y_i) \log (1 - \mathfrak{s}^{(j)}(y = 1 | X_i)) \right] \end{aligned} \quad (9)$$

Where $\mathfrak{W}^{(j)} = \{\mathbb{W}^{(1)}, \mathbb{W}^{(2)}, \dots, \mathbb{W}^{(J)}\}$ is constructed by concatenating the convolutional weights of j hidden layers. Thus, the objective function for branch classifier j can be mathematically expressed as:

$$\mathcal{F}^{(j)}(\mathfrak{W}^{(j)}, \delta_{full}^{(j)}) = \min_{\mathfrak{W}^{(j)}, \delta_{full}^{(j)}} \left(\frac{1}{n} \text{Loss}^{(j)}(\mathfrak{W}^{(j)}, \delta_{full}^{(j)}) \right) \quad (10)$$

Lastly, by aggregating the objective functions of each branch classifier, the objective function of our technique is derived as

$$\mathcal{F}(\mathfrak{W}, \delta_{full}) = \min_{\mathfrak{W}, \delta_{full}} \left(\sum_{j=1}^J \gamma^{(j)} \mathcal{F}^{(j)}(\mathfrak{W}^{(j)}, \delta_{full}^{(j)}) \right) \quad (11)$$

where $\gamma^{(m)}$ denotes the j -th branch classifier weight, and $\delta_{full} = (\delta_{full}^{(1)}, \delta_{full}^{(2)}, \dots, \delta_{full}^{(M)})$. In the present methodology, all branch classifiers influence the training through the average weight strategy, efficiently minimizing additional computing overhead. Linking extra output layers to all hidden layers of CNN effectively addresses the vanishing gradient problem in gradient-based optimization. Usually, this vanishing gradient problem results from computing the gradient chain rule on diminished values during the backpropagation process. To address the issue, the j -th layer's gradient has been strengthened by amalgamating the gradients of branches from j to J . Consequently, the CNN layer with reduced depth will be the gradient.

2) *DQN Reinforcement Optimized Ensemble*: In ensemble formation, all classifiers do not equally contribute to improving performance; certain classifiers may negatively impact overall performance [12]. Therefore, it is imperative to identify an optimal subset of classifiers. To find the optimal solution, the current study employs RL-based Deep Q-Network (DQN). The rationale for employing DQN is that it uses a neural network (NN) as a function approximator rather than maintaining large Q-tables. This facilitates the efficient estimate of Q-values without requiring massive memory. DQN combines Q -learning with NN to approximate Q -value and enhance decision-making [40]. DQN involves an agent for learning optimal actions a_{best} by estimating Q -values, which signify the anticipated cumulative reward R for each action inside a specific state s . Furthermore, in the present study, DQN utilizes the F1 score of classifiers to evaluate the ensemble. The rationale for employing the maximum F1

score is that it balances the trade-off between false positives and false negatives by harmonizing precision-recall, making it effective for skewed distributions.

In the proposed ensemble, the number of classifiers is not fixed. Instead, the study employed an RL to identify the best view-classifier combination for ensemble formation. In our model, $2^{36} - 1$ distinct view and branch classifier combinations are possible. To determine the optimal view classifier pair from a vast search space, DQN employs an agent that interacts with an environment (here, environments represent branch 36 classifiers). To represent the 36 branch classifiers of our ensemble model, we utilize a 36-bit binary encoded string, where each bit indicates the existence or absence of the corresponding classifier in the ensemble. In DQN learning, initially, it generates a random 36-bit binary string representing a state s_t as $s_t = [b^1, b^2, \dots, b^{36}]$. Where $b_i \in \{0, 1\}$ indicates the presence or removal of the i -th classifier in the ensemble construction. For instance, Fig. 1 contains a random state (36-bit string) containing six ones (from CNN-1, 2 & 6), which signifies the selection of six branch classifiers for ensemble construction. Additionally, action $a \in \{1, 2, 3, \dots, 36\}$ decides the inclusion or exclusion of a particular classifier. To select the initial and subsequent actions, an epsilon-greedy policy is utilized that mixes both random actions (exploration) and acquired knowledge (exploitation) during training but with varying probabilities, as mentioned in the equation.

$$a = \begin{cases} \text{random action} & \text{with probability } \epsilon \\ \arg \max_a Q(s_t, a) & \text{with probability } 1 - \epsilon \end{cases} \quad (12)$$

Where the agent selects a random action with probability ϵ , which decreases over time and probability $1 - \epsilon$, increases over time, the agent chooses the best action a_{best} that maximizes the predicted Q - value (i.e., the current best-known action) as

$$a_{best} = \arg \max_a Q(s_t, a) \quad (13)$$

In RL, the agent's objective is to learn a Q - value $Q(s_t, a)$ that calculates the cumulative reward $R(s_t, a)$ for choosing action a in state s_t . Here, the reward R is referred to as the mean F1 score of the chosen classifiers:

$$R(s_t, a) = \frac{\sum_{i=1}^{36} s_t^i \cdot F1_i}{\sum_{i=1}^{36} s_t^i} \quad (14)$$

Where $F1_i$ is the i -th classifier F1 score, while s_t^i signifies the inclusion status of the classifier in the current ensemble. After calculating the reward, the agent decides the next state s'_t and updates Q - values using the Bellman equation as:

$$Q(s_t, a) \leftarrow Q(s_t, a) + \eta \left[R(s_t, a) + \delta \max_{a'} Q(s'_t, a') - Q(s_t, a) \right] \quad (15)$$

Where $R(s_t, a)$ denotes the reward received for taking action a , $\max_{a'} Q(s'_t, a')$ indicates the maximum Q - value for the next state s'_t , δ specifies the discount factor, and η is a learning rate. During the training process, an agent stores its experience (s_t, a, R, s'_t) in a replay buffer. The replay buffer is utilized to sample prior experience during the training process. The agent intermittently extracts a batch of experiences from the replay

buffer and employs them to refine the DQN model by minimizing the loss. The loss function is often defined as the mean squared error (MSE) between the predicted and target Q -values [40].

$$L = \frac{1}{N} \sum_{i=1}^N \left[\left(R(s_t, a) + \delta \max_{a'} Q_{target}(s'_t, a') - Q_{policy}(s_t, a) \right)^2 \right] \quad (16)$$

Where N represents the batch size, Q_{target} refers to the target network employed to stabilize training, whereas Q_{policy} denotes the present policy network. During training, the target network Q_{target} weight $\mathbb{W}$ is updated periodically from the policy network weights. This is accomplished by a soft update mechanism:

$$\mathbb{W}_{target} \leftarrow \tau \mathbb{W}_{DQN} + (1 - \tau) \mathbb{W}_{target} \quad (17)$$

Where $\mathbb{W}_{target}$ and $\mathbb{W}_{DQN}$ are the targets and DQN model weights, respectively, and τ is a minor constant (0.001) that regulates the update rate. After the target model update, if DQN agents converge or the ensemble is optimized, terminate the training process; otherwise, again, the agent selects an action (inclusion or removal of specific classifier for ensemble construction) for the next states s' , calculates the reward, and updates the Q - value and $\mathbb{W}_{target}$ iteratively until the DQN agent convergence. Once the agent converges, it identifies the optimal state s^* , that maximizes the reward as

$$\sum_{q=0}^Q \gamma^q R(s_t^q, a^q) \quad (18)$$

where $q \in \{1, 2, 3, \dots, Q\}$ is the number of episodes, and s^* is a 32-bit string signifying the optimal view-classifier combination. By decoding this 32-bit string, which gives the maximum mean F1, the proposed multiview technique identifies the optimal view-branch classifier pair for ensemble construction. In the present study, this optimal ensemble is utilized to predict the peptide anti-diabetic activity by calculating the mean accuracy of its branch classifiers as

$$Ensem_{Output} = \frac{1}{t} \sum (O(c_1), O(c_2), \dots, O(c_t)) \quad (19)$$

where $Ensem_{Output}$ represents the ensemble output, t represents the total number of selected branch-classifiers in an optimal ensemble, and $O(c_i)$ represents the accuracy of i - th branch-classifier in an optimal ensemble. For better understanding, the DQN-based optimization process algorithm is provided in the supplementary file.

Moreover, to validate the ensemble performance, only accuracy is less significant. Therefore, we employed 10-fold cross-validation on the testing data. Cross-validation mitigated bias through iterative training/testing on different data partitions. In addition, an independent dataset is utilized to measure the model's potential on external data. The significance of DQN optimized ensemble is that it selects optimal branch classifiers, where each classifier makes predictions on progressively abstracted intermediate (shallow to deeper) level features. This optimization process selectively discards redundant or less

TABLE III
EVALUATION OF ML CLASSIFIERS WITH PPSC, EBE, AND SBE VIEW ON ADP-FUSE^{Training} DATASET

Dataset	Classifiers	PPSC				EBE				SBE			
		Acc.	Sen.	Spec.	MCC	Acc.	Sen.	Spec.	MCC	Acc.	Sen.	Spec.	MCC
ADPF ₁ ^{Training}	SVM(linear)	87.77	89.56	86.23	0.763	57.60	91.76	44.75	0.309	78.72	71.63	86.41	0.587
	SVM(rbf)	90.73	93.38	88.73	0.821	92.93	88.65	98.21	0.863	93.77	91.34	97.11	0.885
	LogisticRegression	87.16	88.62	86.19	0.743	65.73	68.56	64.36	0.318	74.70	73.44	77.23	0.492
	GradientBoostingClassifier	95.67	97.08	95.04	0.912	96.35	96.13	98.00	0.961	97.26	97.36	97.15	0.957
	RandomForestClassifier	90.58	91.54	90.88	0.816	90.05	81.45	99.00	0.815	91.03	89.45	93.65	0.823
	XGBClassifier	91.70	93.56	90.31	0.911	90.41	83.69	96.65	0.837	92.32	94.27	90.82	0.831
	AdaBoostClassifier	94.07	95.67	93.76	0.883	94.30	93.45	96.66	0.894	96.50	97.03	96.35	0.949
	lightgbm	93.10	95.65	92.34	0.809	87.10	87.54	87.49	0.668	93.52	96.47	90.22	0.807
ADPF ₂ ^{Training}	SVM(linear)	91.29	96.45	79.67	0.684	85.00	98.51	26.98	0.202	87.00	98.45	56.96	0.689
	SVM(rbf)	93.86	98.61	83.56	0.751	98.14	98.24	94.67	0.952	98.00	98.38	94.31	0.954
	LogisticRegression	90.71	96.37	78.55	0.672	71.86	97.45	21.39	0.159	86.00	98.70	52.78	0.654
	GradientBoostingClassifier	98.41	98.41	99.00	0.970	97.86	98.85	95.71	0.959	98.72	99.02	97.16	0.971
	RandomForestClassifier	93.00	98.66	76.45	0.791	91.57	98.23	70.43	0.724	88.57	97.93	60.66	0.723
	XGBClassifier	98.07	99.00	97.11	0.978	98.22	99.00	97.18	0.976	98.57	99.00	97.43	0.983
	AdaBoostClassifier	97.11	98.11	96.75	0.959	97.43	98.44	96.81	0.957	97.66	98.80	95.93	0.960
	lightgbm	97.45	98.03	96.62	0.960	98.13	98.11	97.92	0.969	97.88	98.20	96.81	0.962

*ADPF - ADP-Fuse, *PPSC-Physicochemical Properties & Sequence Composition, *EB-Evolutionary Based, *SB-Sequence-Based, Note: Acc., Sen., and Spec. are reported in percentage (%), while MCC is reported as a coefficient (0–1).

informative classifiers, such as those trained on highly correlated or overlapping features. This implicit feature selection improves both efficiency and predictive performance without requiring explicit dimensionality reduction.

IV. EXPERIMENTS AND RESULT

The structure of the experimental work has three subsections. Subsection A delineates the experimental setup. Subsection B evaluates the proposed model.

A. Experimental Setup

The investigations employed an NVIDIA 1080ti GPU, 128 GB RAM, Python 3.6, and Keras, which serves as a high-end application programming interface (API) built upon TensorFlow. The efficacy of proposed techniques in predicting antidiabetic activity was evaluated using various metrics, including accuracy, sensitivity, specificity, Matthews correlation coefficient (MCC), and F1 score. The accuracy metrics reflect the overall efficacy of the model. Specificity and Sensitivity play vital roles in medical diagnosis, as they provide information about the rates of accurately detecting negative and positive samples. The MCC measure assesses the classification's performance by considering true positives, false positives, true negatives, and false negatives. F1-score is a harmonic mean of precision and recall. The formulas of the utilized metrics are mentioned in the supplemental file. The source code and datasets used in this work are available at: <https://github.com/akumarphd2022cs-maker/DQNCode.git>

B. Model Evaluation

A sequence of extensive experiments has been conducted to evaluate the impact of a DQN-optimized ensemble. The experimental process consisted of three stages: First, machine learning (ML) classifiers were assessed to predict antidiabetic activity. Next, the proposed ensemble was evaluated. After that, the proposed method was compared with the best-performing ML classifier. Subsequently, SHAP analysis, statistical test, and

ablation study were performed. Lastly, the proposed model was compared with state-of-the-art models.

1) *Evaluation of ML Classifiers With Multiview:* The effectiveness of ML classifiers was confirmed by comparing the outcomes. Based on prior studies, SVM(linear, rbf), LogisticRegression, GradientBoost, RandomForest, XGBclassifier, AdaBoost, and lightgbm algorithms were employed for classification. The 10-fold validation findings are presented in Tables III and IV. The reason for applying ML in antidiabetic activity prediction is its high success rate. Moreover, the present study considered 3 distinct views in experimentation: SB, PPSC, and EB. The optimal values of each dataset are indicated and emphasized in bold font. The findings of ML models on Training data presented in Table III lead to the following conclusion:

- The findings of an experiment performed on ADP-Fuse₁^{Training} antidiabetic and non-antidiabetic datasets demonstrate that the AdaBoostClassifier obtains optimal accuracy in conjunction with an SB view.
- The observation on the ADP-Fuse₂^{Training} T1D and T2D ADPs dataset demonstrates that the GradientBoost performs excellently when paired with an SB view.

Additionally, to validate the robustness and generalizability of the classifiers, an external independent ADP-Fuse^{Independent} datasets were utilized. The experimental findings on the independent datasets are presented in Table IV, from which the following conclusion can be drawn:

- The experimental outcome on the ADP-Fuse₁^{Independent} anti-diabetic dataset demonstrates that the lightgbm with SE View performs best, attaining the highest accuracy.
- Observation on the ADP-Fuse₂^{Independent} T1D and T2D ADPs datasets indicates that again lightgbm surpasses other classifiers and obtains the highest accuracy in conjunction with a PPSC view.

Although ML classifiers exhibit satisfactory results even then, performance metrics like accuracy are less significant, as they might be misleading because they do not account for the nuanced trade-offs between sensitivity, specificity, and MCC. The Receiver Operating Characteristic (ROC) curve is utilized to address this issue. The ROC thoroughly evaluates the

TABLE IV
EVALUATION OF ML CLASSIFIERS WITH PPSC, EB, AND SB VIEW ON ADP-FUSE^{Independent} DATASET

Dataset	Classifiers	PPSC				EB				SB			
		Acc.	Sen.	Spec.	MCC	Acc.	Sen.	Spec.	MCC	Acc.	Sen.	Spec.	MCC
ADPF ₁ ^{Independent}	SVM(linear)	84.69	86.71	84.28	0.645	66.67	88.34	51.78	0.388	76.46	64.56	80.89	0.419
	SVM(rbf)	88.23	92.67	87.32	0.727	85.21	75.05	88.23	0.613	86.98	83.54	88.34	0.678
	LogisticRegression	85.31	86.23	85.64	0.657	59.58	58.82	60.87	0.259	72.81	70.56	74.23	0.404
	GradientBoostingClassifier	87.40	90.45	87.66	0.745	82.29	78.63	84.22	0.562	90.31	92.74	82.40	0.767
	RandomForestClassifier	87.08	84.76	88.25	0.677	79.27	66.78	83.34	0.469	87.08	83.56	88.67	0.678
	XGBClassifier	89.38	92.79	89.38	0.748	82.29	80.61	83.87	0.576	89.90	90.47	90.66	0.757
	AdaBoostClassifier	85.62	90.56	84.37	0.664	77.19	78.34	77.63	0.482	88.54	89.53	88.87	0.715
	lightgbm	90.10	92.56	90.59	0.756	85.10	85.67	85.66	0.644	90.52	94.71	89.48	0.773
ADPF ₂ ^{Independent}	SVM(linear)	90.40	92.45	77.18	0.628	88.14	98.89	25.14	0.295	90.96	99.06	32.22	0.504
	SVM(rbf)	92.66	95.78	77.89	0.681	90.96	94.23	68.65	0.605	94.92	98.09	73.23	0.754
	LogisticRegression	90.40	92.75	77.38	0.624	87.57	98.92	20.84	0.262	90.16	98.00	32.65	0.491
	GradientBoostingClassifier	93.79	96.67	77.29	0.723	89.83	94.56	59.54	0.533	90.40	94.43	68.46	0.584
	RandomForestClassifier	90.40	96.44	50.67	0.528	93.22	99.34	50.44	0.652	90.96	99.00	32.45	0.506
	XGBClassifier	94.35	95.67	86.34	0.761	90.96	96.60	55.24	0.555	91.53	93.89	82.25	0.673
	AdaBoostClassifier	90.40	95.66	59.19	0.556	89.83	95.77	55.75	0.513	93.79	95.33	86.56	0.754
	lightgbm	94.92	96.66	86.14	0.786	90.40	95.34	55.89	0.533	92.09	94.06	82.89	0.749

*ADPF - ADP-Fuse, *PPSC-Physicochemical Properties & Sequence Composition, *EB-Evolutionary Based, *SB-Sequence Based, Note:Acc., Sen., and Spec. are reported in percentage (%), while MCC is reported as a coefficient (0–1).

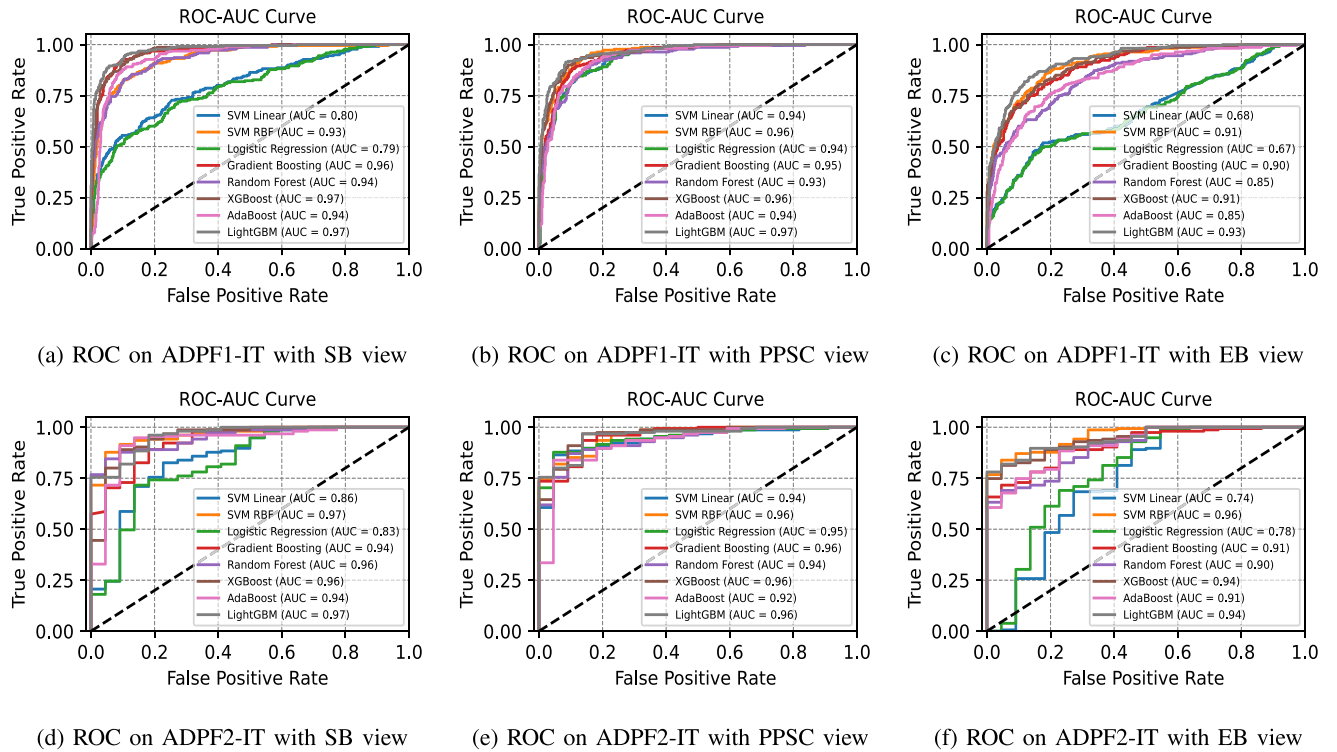


Fig. 3. Performance Analysis of Machine Learning Classifiers using ROC-AUC Curve on Standard ADP-Fuse Dataset. *ADPF1-IT - ADP-Fuse₁^{IndependentTest}, *ADPF2-IT - ADP-Fuse₂^{IndependentTest}, *PPSC-Physicochemical Properties & Sequence Composition, *EB-Evolutionary Based, *SB-Sequence Based.

classifier's discriminative potential at all thresholds by showing the true positive score against the false positive score. To show the ML classifiers' potential, the ROC curve is illustrated in Fig. 3 using an independent dataset with multi-views. In the 1st ADP-Fuse₁^{Independent} antidiabetic dataset, the LightGBM outperforms others, achieving a 0.97 AUC score with both SB and PPSC views. At the same time, the XGBClassifier also performs well, attaining a 0.97 AUC score with the SB view. In the 2nd ADP-Fuse₂^{Independent} T1D and T2D ADPs datasets, the LightGBM and SVM(RBF) surpass other classifiers, attaining a 0.97 AUC score with SB views.

Moreover, the ROC-AUC score of the ML classifier shows the potential of classifiers. However, ROC can also be misleading if the dataset is imbalanced, as the FPR is insensitive to class imbalance, focusing only on negative samples. In the current study, the second ADP-Fuse-training dataset is imbalanced. The Precision-Recall (PR) curve is more appropriate for addressing this issue. PR curve highlights the balance between positive predictive value and sensitivity, which is particularly informative in imbalanced class situations. Fig. 4 illustrates the PR curve on the independent dataset with multi-views. In the first ADP-Fuse₁^{Independent} antidiabetic dataset, the LightGBM

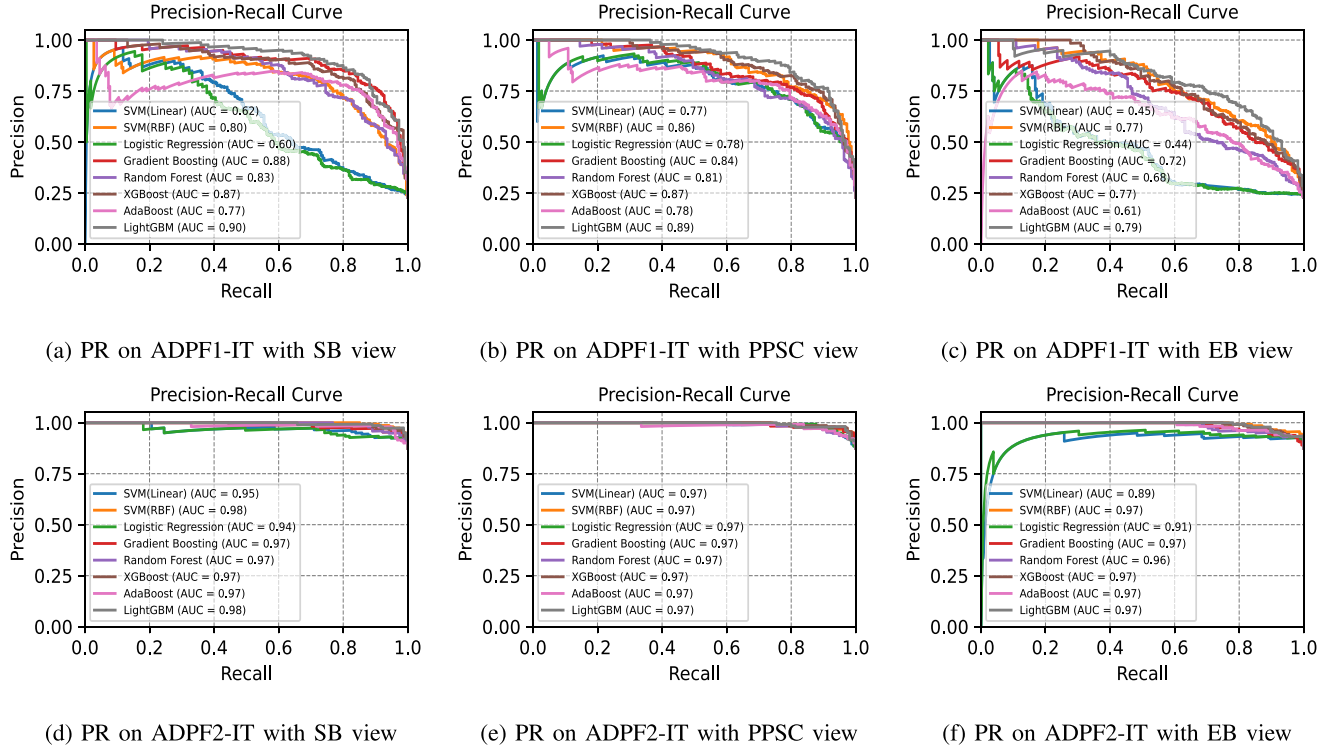


Fig. 4. Performance Analysis of Machine Learning Classifiers using Precision-Recall Curve on Standard ADP-Fuse Dataset. *ADPF1-IT - ADP-Fuse₁^{IndependentTest}, *ADPF2-IT - ADP-Fuse₂^{IndependentTest}, *PPSC-Physicochemical Properties & Sequence Composition, *EB-Evolutionary Based, *SB-Sequence Based.

TABLE V
EVALUATION OF PROPOSED ENSEMBLE ON STANDARD ADP-FUSE DATASET

Dataset	CD-HIT/Without CD-HIT	Acc.(%)	Sen.(%)	Spec.(%)	MCC(0-1)	F1(%)
ADP - Fuse ₁ ^{Training}	CD-HIT	92.65 ± 1.12	93.22 ± 0.54	89.88 ± 2.11	0.860 ± 0.03	86.49 ± 1.94
	Without CD-HIT	95.88 ± 0.97	96.91 ± 0.82	94.88 ± 1.56	0.910 ± 0.03	89.97 ± 2.41
ADP - Fuse ₂ ^{Training}	CD-HIT	95.77 ± 0.92	96.24 ± 1.31	94.06 ± 1.33	0.851 ± 0.02	89.94 ± 1.67
	Without CD-HIT	97.45 ± 0.89	98.10 ± 0.67	96.06 ± 1.51	0.872 ± 0.02	92.11 ± 2.17
ADP - Fuse ₁ ^{IndependentTest}	CD-HIT	92.64 ± 0.78	90.22 ± 1.22	93.41 ± 0.76	0.800 ± 0.05	83.91 ± 2.44
	Without CD-HIT	95.69 ± 0.95	94.57 ± 1.84	95.94 ± 0.89	0.881 ± 0.03	90.10 ± 1.86
ADP - Fuse ₂ ^{IndependentTest}	CD-HIT	94.11 ± 1.03	96.16 ± 0.87	92.45 ± 1.02	0.825 ± 0.02	88.12 ± 0.98
	Without CD-HIT	97.06 ± 0.66	97.45 ± 0.97	95.00 ± 1.11	0.870 ± 0.02	93.08 ± 1.24

with SB view surpasses other classifiers, showing an AUC score of 0.97 on the PR curve. Conversely, in the second ADP-Fuse₂^{IndependentTest} dataset, numerous classifiers demonstrate commendable performance, achieving a 0.98 AUC score. The observed outcomes confirm the model's proficiency in classifying antidiabetic peptides.

2) *Evaluation of Proposed Ensemble*: This study utilized a DQN-optimized ensemble to predict the antidiabetic activity. The experimental outcome with 10-fold validation is presented in Table V. As previously stated in section III-A, earlier studies [11] did not employ CD-HIT for ADPs (applied CD-HIT solely on non-ADPs); they merely eliminated identical ADPs. In contrast, the present study assessed the ensemble by solely eliminating identical ADPs and subsequently applying CD-HIT to the ADPs. The findings presented in the Table V culminate in the conclusion that:

- The investigation on the ADP-Fuse₁^{Training} anti-diabetic dataset reveals that the proposed ensemble without

CD-HIT achieves the maximum classification accuracy of 95.88%. To validate the model's performance, the proposed ensemble was further assessed on the ADP-Fuse₁^{IndependentTest} dataset, achieving a classification accuracy of 95.69% without CD-HIT, and showing 3.05% improvement in accuracy when evaluated with CD-HIT.

- In the ADP-Fuse₂^{Training} dataset for predicting type-1 and type-2 antidiabetic activity, the ensemble without CD-HIT outperforms and shows a classification accuracy of 97.43%. On the other hand, on the ADP-Fuse₂^{IndependentTest} dataset, the proposed ensemble without CD-HIT achieves a 97.06% classification accuracy, representing a 2.95% gain over the ensemble with CD-HIT.

The findings validate the efficacy of the proposed ensemble in classifying antidiabetic activity. Furthermore, as previously discussed in section IV-B1, the ROC and its AUC score are high-influence measures, while the RP curve addresses the challenge of class imbalance, facilitating biological data analysis.

TABLE VI
COMPARISON OF THE PROPOSED ENSEMBLE WITH BEST PERFORMING INDIVIDUAL CLASSIFIER ON STANDARD DATASET

Dataset	Classifiers	View	Acc.(%)	Sen.(%)	Spec.(%)	MCC(0-1)	AUC(0-1)
$ADP - Fuse_1^{IndependentTest}$	Lightgbm	SB	90.52	94.71	89.48	0.773	0.970
	Proposed Ensemble	Multiview	95.69	94.57	95.94	0.872	0.978
$ADP - Fuse_2^{IndependentTest}$	Lightgbm	PPSC	94.92	96.66	86.14	0.786	0.972
	Proposed Ensemble	Multiview	97.06	97.45	95.00	0.870	0.989

*PPSC-Physicochemical Properties & Sequence Composition, *SB-Sequence-Based,

TABLE VII
ENSEMBLE ERROR ANALYSIS

Dataset	TP	TN	FP	FN	Error (%)	FPR (%)	FNR (%)
1	209	709	30	12	4.38	4.06	5.43
2	153	19	1	4	2.82	5.00	2.55

*1- $ADP - Fuse_1^{Independent}$, *2- $ADP - Fuse_2^{Independent}$

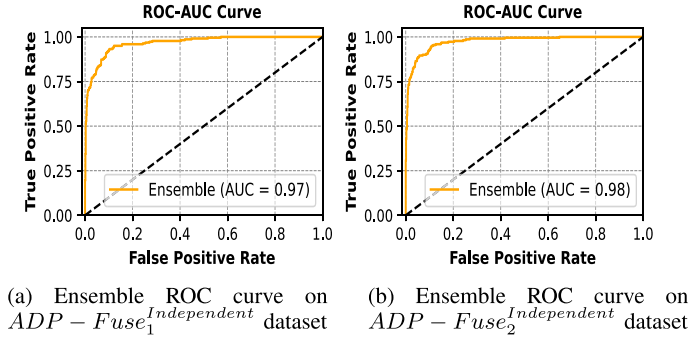


Fig. 5. Analysis of proposed ensemble using ROC curve.

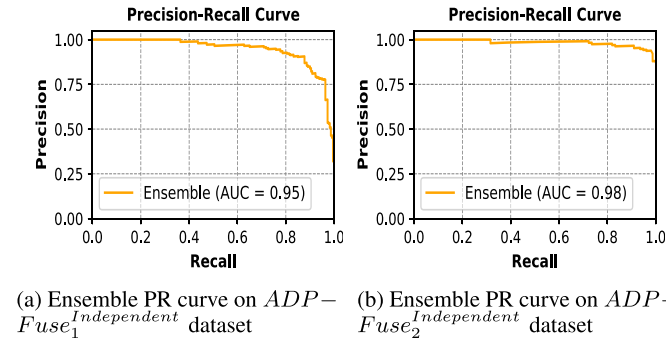


Fig. 6. Analysis of proposed ensemble using PR curve.

The ROC curve depicts the relationship between true positives and false positives. Conversely, the RP curve highlights the balance between sensitivity and positive predictive value. The ROC and PR curves of the proposed ensemble are illustrated in Figs. 5 and 6 on independent datasets, respectively. As illustrated in Fig. 5, on the $ADP - Fuse_1^{Independent}$ antidiabetic dataset, the proposed ensemble achieves a 0.97 AUC score. Similarly, on the $ADP - Fuse_2^{Independent}$ T1D and T2D ADPs dataset, the proposed ensemble also demonstrates the highest AUC score of 0.98. Furthermore, as illustrated in PR curve Fig. 6, the proposed ensemble gets the highest 0.95 AUC score on the $ADP - Fuse_1^{Independent}$ antidiabetic dataset and 0.98 AUC on the $ADP - Fuse_2^{Independent}$ T1D and T2D ADPs dataset. Lastly, an error analysis is conducted to examine the model's limitations, as presented in Table VII. In the first dataset, the ensemble exhibits an error rate of 4.38%, with a false positive rate (FPR) of 4.06% and a false negative rate (FNR) of 5.43%. In a second dataset, the ensemble has an error rate of 2.82%, with a FPR of 5% and an FNR of 2.55%.

Furthermore, the convergence analysis performed by depicting the DQN loss, epsilon decay, and Q-value learning on the training data in Fig. 7. As illustrated in the Fig. 7, DQN loss is computed using mean squared error, and the Q-value is directly proportional to the training F1 score. In the Fig. 7(b)–(e), the epsilon decay for both datasets is identical, as epsilon_decay is 0.995 in each dataset. The validation curve of the ensemble training vs testing F1 score, illustrated in a Fig. 8, demonstrates consistent learning and an absence of overfitting. The reward convergence of the ensemble is depicted in Fig. 9, where the red line represents the highest reward achieved so far, and the blue line indicates the 50-episode moving average. Additionally, we compared the proposed ensemble convergence with that of the random search ensemble. In the $ADP - Fuse_1$ dataset, random search converges rapidly within 293 iterations but attains an accuracy that is 1.93% lower than that of the proposed ensemble. In a $ADP - Fuse_2$ dataset, the proposed ensemble converges faster than random search. The proposed models achieve 32.4% and 15.3% episode efficiency in the $ADP - Fuse_1$ and 2 datasets, respectively. The performance metric comparison between the proposed model and random search is also presented in the ablation study section later. Lastly, to enhance the comprehension of the optimal multiview learning ensemble, we conducted a Shapley Additive Explanations (SHAP) analysis. SHAP value measures the impact of each view-classifier pair in ensemble performance (F1-score) and is illustrated in Fig. 10. As illustrated in Fig. 10(a), a total of 9 view-classifier pairs were selected in an optimal ensemble to predict anti-diabetic activity on the $ADP - Fuse_1^{Independent}$ dataset. Branch-classifier1 of CNN4, with a PPSC view(CNN4-BC1-PPSC), made a significant contribution to the ensemble, whereas branch-classifier3 of CNN5, with an SBE view(CNN5-BC3-SBE), had the lowest contribution. Similarly, Fig. 10(b) illustrates the impact of the selected view-classifier pair on the ensemble, as assessed by SHAP on the T1D and T2D $ADP - Fuse_2^{Independent}$ dataset. A more detailed explanation of SHAP is provided in the supplemental file.

3) *Comparison Between Proposed Technique and Best Performing ML Classifier:* The performance metrics of the best-performing classifier and the proposed ensemble with multiviews are contrasted in Table VI. Different datasets are displayed in rows, whereas the combination of columns 2 and 3

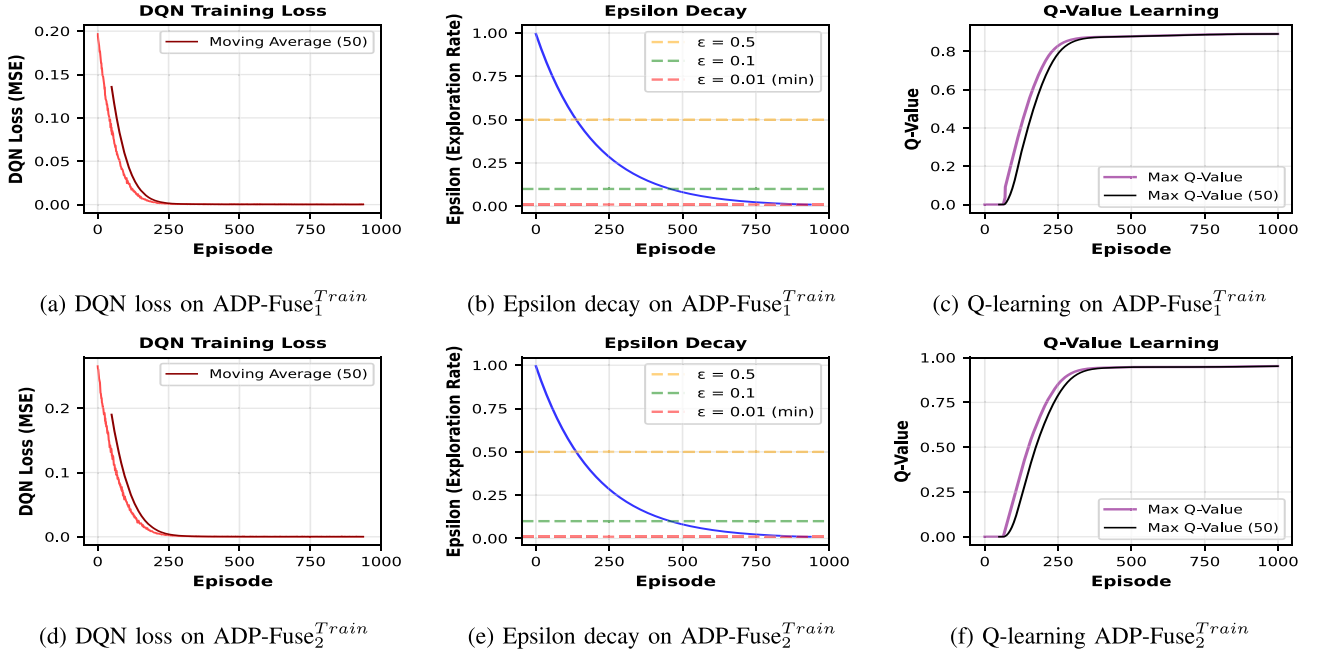


Fig. 7. Convergence analysis of the DQN optimization using DQN-loss, epsilon decay, and Q-value learning.

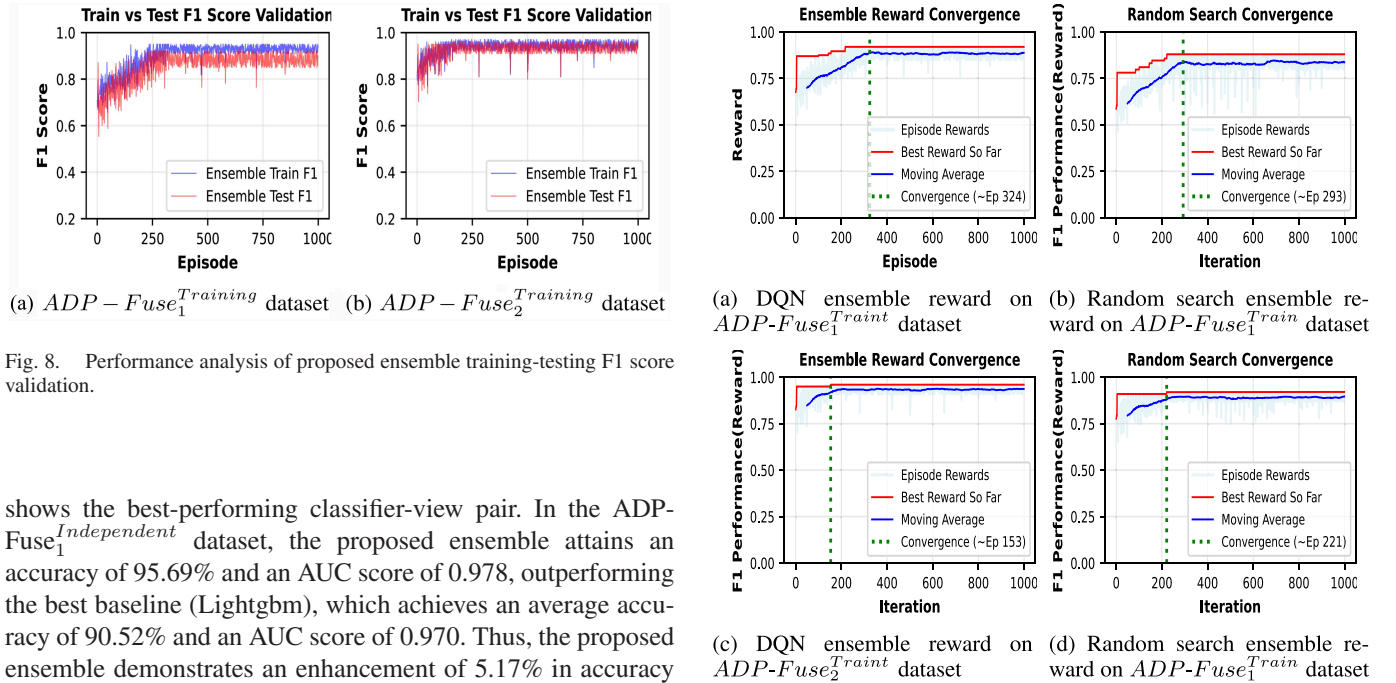


Fig. 8. Performance analysis of proposed ensemble training-testing F1 score validation.

shows the best-performing classifier-view pair. In the $ADP-Fuse_1^{Independent}$ dataset, the proposed ensemble attains an accuracy of 95.69% and an AUC score of 0.978, outperforming the best baseline (Lightgbm), which achieves an average accuracy of 90.52% and an AUC score of 0.970. Thus, the proposed ensemble demonstrates an enhancement of 5.17% in accuracy and 0.008 in AUC score over the best baseline classifier. Next, the proposed ensemble attains 97.06% accuracy and a 0.989 AUC score on the $ADP-Fuse_2^{Independent}$ dataset, reflecting enhancements of 2.14% in accuracy and 0.017 in AUC score relative to the top-performing Lightgbm classifier. The proposed ensemble outperforms the best-performing ML classifier in both datasets, which confirms the ensemble's potential in classification.

4) *Non-Parametric Statistical Test*: To validate the potential of the proposed model, we conducted a non-parametric Wilcoxon signed-rank test. Table VIII presents the outcome of the Wilcoxon test, where the second column represents the

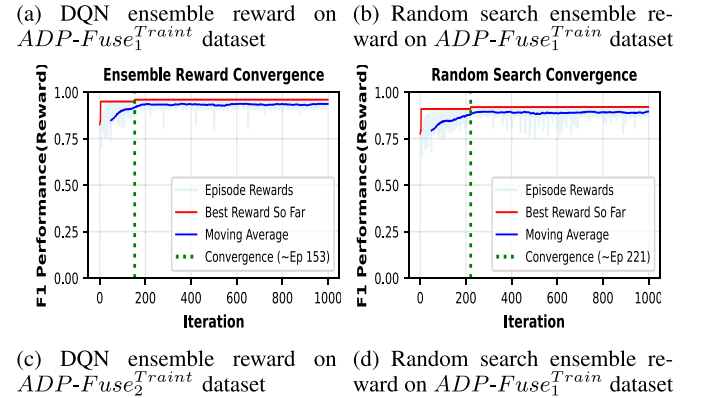


Fig. 9. Reward convergence of proposed DQN optimized ensemble vs random search ensemble on training dataset.

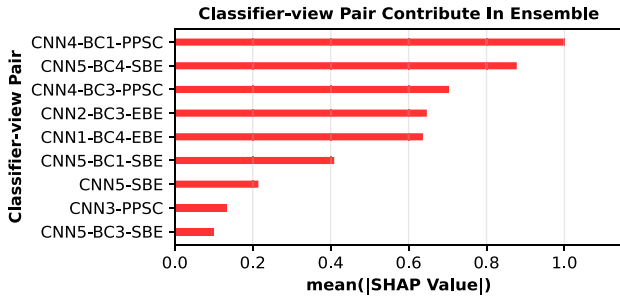
TABLE VIII
WILCOXON STATISTICAL TEST ANALYSIS

Model	Average rank	p-value	Win-Tie-Loss		
			+	±	-
AntiDMPpred	3.75	0.2670	2	0	0
STADIP	2.75	0.2500	2	0	0
ADP-Fuse	2.50	0.0525	4	0	0
Proposed Ensemble	1.00				

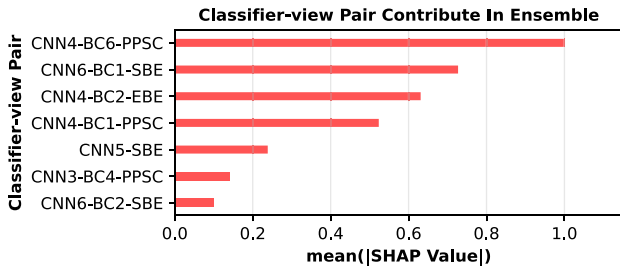
TABLE IX
ABLATION STUDY: IMPACT OF DQN OPTIMIZATION, MULTIVIEW LEARNING, AND ENSEMBLE TECHNIQUES

DS	DQN vs without DQN					Single View vs Multiview					Other Technique vs DQN					
	Methods	Acc	Spec.	Sens.	MCC	Views	Acc	Spec.	Sens.	MCC	Ensemble	Acc	Spec.	Sens.	MCC	SM
1	Best Single BC★	93.22	94.52	92.46	0.840	EBE	90.26	91.51	92.53	0.800	Random Search	93.91	95.66	92.87	0.856	12
	Simple Averaging★	92.77	93.76	91.22	0.835	PPSC	93.08	94.60	92.06	0.839	AutoML	95.06	96.43	94.22	0.877	7
	Simple Voting★	93.06	94.37	91.97	0.839	SBE	93.46	95.33	91.16	0.841	Proposed DQN	95.69	94.57	95.94	0.871	9
	Proposed DQN	95.69	94.57	95.94	0.871	Multiview	95.69	94.57	95.94	0.871						
2	Best Single BC★	95.45	97.22	93.08	0.879	EBE	92.66	91.51	93.56	0.798	Random Search	95.78	95.88	94.76	0.880	10
	Simple Averaging★	94.21	95.33	92.50	0.877	PPSC	93.52	94.96	91.68	0.840	AutoML	96.22	97.23	95.89	0.882	8
	Simple Voting★	95.88	96.67	94.11	0.880	SBE	95.14	95.83	94.15	0.866	Proposed DQN	97.06	97.45	95.00	0.875	7
	Proposed DQN	97.06	97.45	95.00	0.875	Multiview	97.06	97.45	95.00	0.875						

*DS-Dataset, *1-ADP-Fuse₁^{Independent}, *2-ADP-Fuse₂^{Independent}, *★-without DQN, *BC-Branch Classifier, *SM-Selected Models *PPSC-Physicochemical Properties & Sequence Composition, *EB-Evolutionary Based, *SB-Sequence Based, Note: Acc., Sen., and Spec. are reported in percentage (%), while MCC is reported as a coefficient (0–1).



(a) Ensemble SHAP analysis on $ADP - Fuse_1^{Independent}$ dataset



(b) Ensemble SHAP analysis on $ADP - Fuse_2^{Independent}$ dataset

Fig. 10. View–classifier contribution analysis using SHAP.

average rank achieved by each method. Findings demonstrate that the proposed model achieved the highest (1.0) average rank. In addition, the win-tie-loss statistics reveal that the proposed models win in 4 datasets and outperform ADP-Fuse. Similarly, the proposed model wins in 2 datasets, surpassing both STADIP and AntiDMPpred. In addition, as detailed in the third column, the proposed model is significant at (0.05) as compared to ADP-Fuse. Although when compared with STADIP and AntiDMPpred, the achieved p-value ($p > 0.05$) indicates that the proposed model is not significant, the highest average rank and positive sign counts present substantial evidence of superiority. A more detailed explanation of the Wilcoxon test and confidence interval is provided in the supplemental file.

5) *Ablation Study*: To thoroughly analyze the significance of multiview, DQN optimization, and ensemble performance, we conducted an ablation study. This study was organized into three blocks, presented in Table IX. First, under DQN vs.

without DQN, the proposed DQN optimized ensemble consistently outperforms without DQN optimized approaches (best single branch-classifier(BC), simple averaging ensemble, and voting ensemble) across both datasets, achieving accuracies of 95.69% and 97.06%, respectively. Second, in a single vs. multiview comparison, the proposed ensemble with single EBE, PPSC, and SBE views achieves accuracies of 92.26%, 93.08%, and 93.46% in dataset-1 and 92.66%, 93.52%, and 95.14% in dataset-2, respectively. In contrast, the proposed ensemble with multiview outperforms all single views and shows a 2.2%(in dataset-1) and 1.9%(in dataset-2) improvement over the best-performing single(SBE) view. Lastly, in other techniques vs. DQN, the proposed DQN is evaluated against random search and AutoML ensemble. Here, random search explored model combination using a random search strategy, while the AutoML framework employed a greedy search, sequentially choosing the most promising combination based on performance feedback. For a fair comparison, both techniques also evaluated the ensemble that utilizes the F1 score, running 1000 iterations/epoch with a step_size = 1. As detailed in Table IX, DQN exhibits a marginal 0.63%(in dataset-1) and 0.84%(in dataset-2) improvement over the other best-performing AutoML technique and outperforms it. Finally, the last column, SM (selected models), indicates the number of models selected for the optimal ensemble construction. The findings presented in Table IX highlight the significance of multiview and DQN optimization in ensemble performance.

6) *Compare the Proposed Technique to State-of-The-Art*: To illustrate the usefulness of the proposed model, it is compared to other state-of-the-art models in Table X. To the best of our knowledge, only two predictors, AntiDMPpred [10] and ADP-Fuse [11], currently exist for predicting ADPs. The AntiDMPpred exclusively predicts ADP, while ADP-Fuse predicts both ADP and its type. Therefore, AntiDMPpred was compared only on the ADP-Fuse1 dataset. Each technique was assessed and contrasted utilizing a 10-fold validation approach. While predicting peptides antidiabetic activity on the ADP-Fuse₁^{Independent} dataset, the proposed ensemble outperforms the existing techniques and demonstrates 95.72% classification accuracy and 0.988 AUC score. Moreover, in T1D and T2D ADPs prediction on the ADP-Fuse₂^{Independent} dataset, the proposed model demonstrates significant enhancement, attaining a 97.00% classification accuracy and 0.995 AUC score. Lastly,

TABLE X
COMPARATIVE ANALYSIS OF DIFFERENT STATE-OF-THE-ART METHODS FOR PREDICTING ANTI-DIABETIC ACTIVITY AND THEIR TYPES

Dataset	Classifiers	Acc.(%)	Sen.(%)	Spec.(%)	MCC(0-1)	AUC(0-1)
$ADP - Fuse^1_{IndependentTest}$	AntiDMPpred	57.0	73.2	52.1	0.214	0.734
	STADIP	95.40	95.30	95.90	0.877	0.990
	ADP-Fuse	94.00	94.50	93.80	0.841	0.987
	Proposed Ensemble	95.69	94.57	94.94	0.871	0.978
$ADP - Fuse^2_{IndependentTest}$	ADP-Fuse	96.6	96.8	95.4	0.858	0.993
	Proposed Ensemble	97.06	97.45	95.00	0.870	0.989

the findings presented in Table X confirm the proposed model's potential in antidiabetic activity prediction.

V. CONCLUSION AND FUTURE SCOPE

This work presents a novel ensemble integrating a multi-view feature learning to address peptide antidiabetic activity prediction. Typically, peptide function or structure prediction commences with amino acid encoding, a fundamental pillar of successful study. However, not all features(views) give the best performance with all classifiers. Therefore, the identification of an optimal view-classifier pair is essential. Consequently, we conducted a systematic classification of different encoding methods. According to the information sources and extraction approach, these encoding are categorized into three groups, each regarded as a distinct view. These views are based on sequence, physicochemical characteristics with sequence composition, and evolution-based views. The performance of each view is regarded as an indicator of its efficacy in antidiabetic activity classification. Additionally, to predict antidiabetic activity, the proposed ensemble effectively selects the optimal view-classifier pair using DQN, which is not achievable by traditional classifiers. To determine the optimal view-branch classifier, RL utilized the F1 score. For experimental purposes, four benchmark datasets were used. The extensive testing results confirm the proposed model's potential. The highest accuracy and AUC score obtained with the optimal multiview in all benchmark datasets show the model's efficacy compared to other models. The main reason for achieving the highest AUC score is the DQN-based optimization for selecting the optimal view-branch classifier pair. Moreover, further investigations on alternative optimal multiview learning techniques point to a fascinating avenue for future investigation.

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